

Signal-Adaptive GraphRAG for Traditional Chinese Bible Question Answering: End-to-End Architecture, Knowledge-Graph Quality Remediation, and Rank Fusion

Ko-Han Chung, Yoke-Hou Tan

Abstract—Domain-specific question answering (QA) with large language models (LLMs) faces high API cost, data-privacy constraints, and strict factuality requirements. We present the complete design, deployment, and three-round empirical optimization of a graph-based retrieval-augmented generation (GraphRAG) system for Traditional Chinese Bible QA. The offline pipeline converts 66 books of the Chinese Union Version from PDF into a pericope-centric hierarchical corpus by layout-geometry analysis, applies token-aware hierarchical chunking, and builds a knowledge graph with grounded dual-track entity extraction (9,120 entities of six types; 173,896 mentions), closed-schema grounded relation extraction (37 relation types), and a 250,418-edge cross-reference network that integrates the Treasury of Scripture Knowledge. Content, vectors, and graph are stored in PostgreSQL, Qdrant, and Neo4j, and served online by a signal-driven six-route retrieval architecture with eleven strategies, cross-encoder reranking, and a linear rank-fusion layer $s_{\text{fused}} = (1-\alpha)s_{\text{rr}} + \alpha w$. On a 100-question benchmark spanning five question types we report a three-round evidence chain: (i) signal-adaptive retrieval beats a semantic-only baseline by +15 percentage points hit rate, independently of the answer model; (ii) a knowledge-graph completeness overhaul (entity anchor coverage 83.8% $\rightarrow$ 98.2%, cross-references $\times 273$) yields *zero* retrieval gain on the head-entity benchmark — a negative result we diagnose, question by question, to a reranker-monopolized final ranking, topical cross-reference noise, and benchmark head saturation; and (iii) a one-day rank-fusion remediation recovers and surpasses all prior results (hit rate 0.97, event-narrative hit rate 1.00, +0.16 over the semantic baseline), with an ablation showing a complementary structure between discrete data repairs and continuous score fusion. We conclude that knowledge-graph construction quality, the retrieval ranking mechanism, and benchmark design *jointly* determine observable benefit, and that optimization order should follow the bottleneck layer rather than the upstream-first order of the data flow.

Index Terms—Evaluation methodology, GraphRAG, knowledge graph construction, negative results, question answering, rank fusion, retrieval-augmented generation, Traditional Chinese.

I. INTRODUCTION

LARGE language models (LLMs) excel at open-domain question answering, but deploying them for a specific knowledge domain exposes three persistent obstacles. First,

commercial API pricing places a recurring cost on every query, which is prohibitive for resource-constrained organizations such as churches, seminaries, and non-profit publishers. Second, sensitive corpora and user questions often cannot leave the premises, ruling out cloud-only inference. Third, and most specific to the religious domain, factual grounding is non-negotiable: recent measurements report that even state-of-the-art LLMs score lowest on faith-related dimensions of a human-flourishing alignment benchmark [1], reinforcing a legitimate concern in the Christian community about whether LLM answers are actually grounded in Scripture rather than in the model’s parametric memory.

Retrieval-augmented generation (RAG) [2] mitigates factual errors by conditioning generation on retrieved evidence, and graph-based RAG (GraphRAG) variants [3], [4] additionally exploit entity and relation structure for multi-hop and aggregation queries. However, most published GraphRAG systems are evaluated as end-to-end black boxes: when a knowledge graph (KG) improvement does or does not move the final metric, the community rarely learns *which layer* — extraction, storage, retrieval, ranking, generation, or the benchmark itself — gated the outcome. This paper documents a deployed system in which exactly that question was forced into the open, three times, by concrete engineering incidents.

We design, build, deploy, and iteratively optimize a GraphRAG system for question answering over the Traditional Chinese Bible (Chinese Union Version, new punctuation edition; CUV). The system converts 66 books from PDF into a pericope-centric hierarchical corpus, constructs a knowledge graph with grounded entity and relation extraction, indexes three complementary databases (PostgreSQL, Qdrant, Neo4j), and serves queries through a signal-driven six-route retrieval architecture terminated by a cross-encoder reranker and a linear rank-fusion layer. Along the way we performed a quantified audit of KG construction quality, executed a completeness overhaul, observed a rigorously verified *negative result* — the overhaul produced zero retrieval improvement on our benchmark — diagnosed the mechanism question by question, and repaired the actual bottleneck (the ranking layer) in one day of engineering, surpassing all previous results.

This paper makes six contributions:

- 1) **A complete, reproducible pipeline from raw PDF to GraphRAG for Traditional Chinese biblical text** (Sections III and IV): layout-geometry PDF parsing

K.-H. Chung and Y.-H. Tan are with the Department of Intelligent Computing and Big Data, Chung Yuan Christian University, Taoyuan City, Taiwan (corresponding author: Y.-H. Tan).

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without OCR or LLMs, pericope-centric hierarchical chunking with the embedding model’s own tokenizer, grounded dual-track entity extraction (six types, 9,120 entities, 173,896 mentions), closed-schema grounded relation extraction (37 relation types with evidence-span verification), and a three-source cross-reference network of 250,418 edges integrating the public-domain Treasury of Scripture Knowledge (TSK) [5].

- 2) **A signal-adaptive six-route retrieval architecture over a triple-database backend, terminated by a rank-fusion layer** (Section V): six boolean query signals select among six routes plus a fallback; each route runs a weighted combination of eleven retrieval strategies; candidates are deduplicated, reranked by a cross-encoder, fused by $s_{\text{fused}} = (1 - \alpha) s_{\text{rr}} + \alpha w$, and finally adjusted by a small set of “user-literal” pins.
- 3) **A quantified causal chain from chunking granularity to KG quality** (Section VIII): we show that the damage to KG quality was caused not by chunk-size parameters but by the *silent inheritance* of an embedding-oriented granularity structure by the extraction, import, and retrieval layers (mechanisms M1–M6), producing a 55.9% silent mention loss and 1,474 retrieval-blind entities. The finding is aligned with, and locally re-validates, published evidence on extraction granularity [3], document-level relation extraction [6], and coreference-resolved KG construction [7]–[9].
- 4) **A verified negative result with a mechanism-level diagnosis** (Section IX): a KG completeness overhaul that raised entity anchor coverage from 83.8% to 98.2% and multiplied cross-references by 273 produced *no* improvement — and locally a regression — on a 100-question head-entity benchmark. Question-by-question forensics attribute this to (a) topical cross-reference votes acting as noise on event-narrative questions, (b) a final ranking monopolized by the cross-encoder’s surface-form score, and (c) benchmark head saturation. This result complements benchmark-level observations that KG defects propagate non-linearly to answers [10] and that GraphRAG gains are frequently over-read due to evaluation bias [11].
- 5) **A rank-fusion remediation with an interpretable ablation** (Section X): TSK de-noising by vote-weighted edge declassing, a curated head-event backfill that re-enabled entity-vector retrieval, and the linear fusion layer together lift hit rate to 0.97 (event-narrative questions to 1.00), surpassing the pre-overhaul state and widening the margin over the semantic-only baseline to +0.16. The α ablation exposes a complementary structure — discrete data repairs saturate hit rate, continuous fusion recovers tail anchors — and a mechanism-level finding that discrete “pin” patches flip from net-positive to net-negative once continuous fusion exists.
- 6) **A methodological conclusion for GraphRAG practice** (Section XI): KG construction quality, the retrieval ranking mechanism, and benchmark design jointly determine the observable benefit of any single-layer investment. In our deployment, one day of ranking-layer repair outper-

formed what a week-scale re-extraction was projected to deliver on the same benchmark; three independent incident rounds show graph assets whose value was gated by retrieval-side mechanisms. We argue that evaluation design is a *prerequisite* for — not an afterthought of — further construction-side optimization.

The remainder of the paper follows the pipeline order: corpus and preprocessing (Section III), knowledge-graph construction (Section IV), online retrieval and generation (Section V), the evaluation framework (Section VI), the quality audit and the three experiment rounds (Sections VII–X), discussion (Section XI), limitations and future work (Sections XII–XIII), and conclusion (Section XIV). Appendix A maps each pipeline stage to the literature it draws on.

II. RELATED WORK

A. Retrieval-Augmented Generation and Hybrid Retrieval

RAG [2] couples a parametric generator with non-parametric retrieval; surveys [12] catalogue its now-extensive design space. Dense passage retrieval [13] and lexical scoring [14] remain the two complementary retrieval substrates, commonly combined by reciprocal rank fusion (RRF) [15]. Our system uses BGE-M3 [16] for dense embeddings (and its companion cross-encoder, BGE-reranker-v2-m3, for reranking), BM25 with Chinese-specific CKIP tokenization [17] for sparse vectors, and RRF for dense-sparse hybrid search. Distinct from standard hybrid search, the final ranking in our system linearly fuses the cross-encoder score with a *strategy-prior* weight that encodes which retrieval strategy proposed the candidate — a channel by which graph evidence survives into the final top- k (Section V-F). Calibrated fusion of heterogeneous graph and vector scores has recently been studied in multi-hop QA [18]; our fusion is deliberately simpler (a single linear interpolation with one global α) and is evaluated by ablation.

B. Graph-Based RAG

GraphRAG [3] popularized LLM-built entity graphs with hierarchical community summaries for query-focused summarization; subsequent systems explore lighter indexing and different retrieval operators, e.g., LightRAG’s dual-level keyword retrieval [19], personalized-PageRank retrieval over passage-linked graphs [20], attributed-community hierarchical retrieval [21], and recursive abstractive summary trees (RAPTOR) [22]. Event-centric variants build explicit event nodes and temporal edges [23], [24]. Benchmark studies ask *when* graphs actually help RAG [25]. Our architecture differs from these retrieval-operator designs in that graph traversal is one of ten strategies selected *per query* by boolean signals (route-level adaptivity), and in that the paper’s empirical focus is not a new operator but the interaction between graph asset quality, ranking, and benchmark — an interaction that, to our knowledge, has not been documented at incident-level granularity for a deployed GraphRAG system.

C. LLM-Based Knowledge Graph Construction

LLM-driven KG construction is surveyed in [26], which contrasts schema-based and schema-free extraction paradigms; our relation extractor is schema-based with a closed 37-type ontology. On construction quality, GraphRAG’s own ablation shows chunk granularity directly controls extraction density (600- vs. 2,400-token chunks yield nearly twice the entity references) [3]; DocRED quantifies that at least 40.7% of relational facts require cross-sentence evidence and 17.6% require coreference reasoning [6]; and “lost-in-the-middle” effects bound how much context can be packed into a single extraction call [27]. Coreference-resolved construction reduces node duplication by 28–45% in the CORE-KG / LINK-KG line of work [7]–[9], with the ablation in [8] cleanly separating coreference (deduplication) from structured prompting (noise suppression). Post-extraction correction is an alternative to re-extraction: ontology-grounded repair [28], canonicalization after open extraction [29], KG denoising that *shrinks* graphs yet improves downstream RAG [30], retrieval-augmented construction that re-retrieves cross-chunk context for each pre-entity [31], restoring the source context of isolated triples [32], and GNN-guided cross-chunk graph augmentation [33]. Atomic-fact decomposition with dual-time modeling [34] shows finer construction units increase fact coverage. Our audit (Section VIII) maps each of our locally measured defects onto this literature, and our remediation roadmap adopts its recipes.

D. Evaluating RAG and GraphRAG

We evaluate generation with RAGAS [35] plus a custom LLM-judged answer-coverage metric, and retrieval with deterministic ground-truth metrics. Two recent results shape our methodology. BRINK benchmarks how KG-based RAG breaks under incomplete knowledge and finds that answer quality degrades non-linearly with KG defects, with systems partly relying on parametric memory rather than structural reasoning [10]. An unbiased evaluation framework for GraphRAG shows that after removing position and verbosity bias from LLM-judged comparisons, reported win rates can drop by tens of points (e.g., LightRAG 66.70% $\rightarrow$ 39.06%) [11]. Our own evaluation encountered a concrete instance of judge bias — the RAGAS answer-relevancy “noncommittal” rule zeroes honest refusals, deflating the score of a cautious commercial model (Section VI-B). Chunking-strategy sensitivity for *retrieval* (not construction) has been studied systematically in [36]; we cite it only for its retrieval-side scope, since no peer-reviewed quantification exists for overlap effects on *extraction*, and we rely on DocRED [6] as indirect evidence there.

III. CORPUS AND DATA PREPROCESSING

The offline pipeline (Fig. 1) turns 66 books of the CUV from PDF into (i) a hierarchical structured corpus, (ii) a multi-granular embedding queue, and (iii) the inputs of knowledge-graph construction. All build steps run on the host from a dedicated `scripts/` environment; every intermediate artifact is materialized as JSONL before database import, so each stage is independently inspectable and re-runnable.

A. Source Corpus

The source is the Chinese Union Version with new punctuation, obtained as 66 per-book PDFs (one per biblical book). The CUV is the de facto standard Protestant translation in the Traditional Chinese world; its vocabulary is early-20th-century literary Chinese, which creates a systematic lexical gap between modern user phrasing and corpus surface forms — a gap that recurs throughout this paper (e.g., the modern phrase “the Last Supper” never occurs in the CUV text, which narrates a “Passover meal”).

B. PDF-to-Markdown by Layout Geometry

Conversion (`convert_bible_pdf.py`, PyMuPDF) uses *pure layout geometry* over character-level spans — no OCR and no LLM — so the step is deterministic, free, and auditable:

- font size ≤ 8.5 pt: running headers/footers, dropped;
- font size ≥ 16 pt: book title or chapter number;
- font size = 12 pt with $x \geq 85$: pericope (section) heading;
- font size ≤ 9.5 pt and numeric: verse number;
- verse number at $x < 75$ with line right-edge < 300 : poetry (line breaks preserved); otherwise prose (lines joined);
- bold N:N tokens: footnotes, aggregated per chapter.

The resulting Markdown uses # for book, ## for chapter, ### for pericope headings, and bold verse numbers. Under each pericope heading, the CUV’s printed parallel-passage annotations (e.g., “(Mark 1:9–11; Luke 3:21–22)” under the baptism pericope) are retained — they become the first source of the cross-reference network (Section IV-C).

C. Pericope-Centric Hierarchical Structure

A *pericope* is the established interpretive unit of biblical scholarship: a self-contained narrative episode, discourse, or psalm. The CUV’s editorial pericope headings give us this segmentation for free, and the system adopts the pericope as the core unit for embedding, entity anchoring, and cross-referencing, for three reasons:

- 1) **Semantic completeness.** A story is never split down the middle; retrieval returns interpretable wholes.
- 2) **Embedding fit.** The median pericope is 388 characters (P90 = 931, max = 3,544), well inside the effective input range of the BGE-M3 embedder [16].
- 3) **Explainability.** Returning “Isaac takes a wife (Gen. 24)” is more interpretable to end users than a verse-range span.

Parsing (`process_bible.py`) with `markdown_parser.py` yields the hierarchy Book $\rightarrow$ Chapter $\rightarrow$ Pericope with counts 66/1,189/2,779, plus per-pericope verse spans. A regex-based reference parser with a Chinese book-abbreviation table resolves the parallel-passage annotations; every reference is *upgraded to pericope level* (never verse level), keeping the cross-reference network aligned with the retrieval unit. Hierarchical IDs are colon-delimited (`gen  $\rightarrow$  gen:1  $\rightarrow$  gen:1:0` for the first pericope of Genesis 1), a scheme reused verbatim across all three databases.

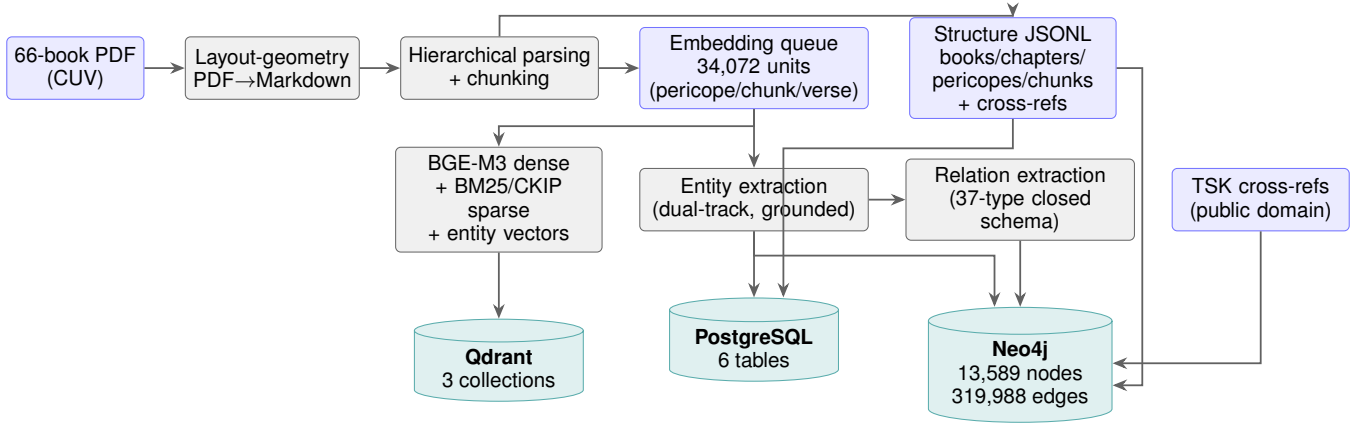


Fig. 1. Offline build pipeline. Every arrow is a materialized JSONL hand-off; the three databases play complementary roles (Section V-A); database counts show the live post-remediation state (July 6, 2026). The embedding queue doubles as the extraction feed — the structural root cause analyzed in Section VIII (mechanism M1).

D. Hierarchical Chunking

Long pericopes are split for embedding by a token-aware hierarchical chunker whose tokenizer is *BGE-M3*’s own, i.e., the chunking layer is explicitly designed for the embedding model [16]:

- pericopes with ≤ 768 tokens (93.9%) are kept whole;
- longer pericopes are split on *verse boundaries* targeting 512 tokens per chunk, with a 1-verse overlap between adjacent chunks;
- a trailing chunk under 128 tokens and ≤ 3 verses is merged back into its predecessor;
- chunk IDs extend the parent ID ($gen:1:0:0$), preserving the hierarchy.

Only 169 of 2,779 pericopes (6.1%) are split, producing 431 chunks; the split pericopes concentrate in Old Testament narrative books (Leviticus 13, Judges 12, 1 Samuel 12, Genesis 11, Numbers 11, Joshua 10, Deuteronomy 9, 2 Samuel 9) and account for 18.3% of the corpus text (243,773 of 1,329,046 characters); their median length is 1,353 characters (max 3,544), against a chunk median of 644 (max 796). Each retrieval unit’s embedded text carries a locating prefix — “*Book, Chapter N, pericope title (verses X–Y):*” — so vectors encode provenance context. Table I summarizes the corpus.

E. Multi-Granularity Embedding Queue

Retrieval needs different granularities for different question shapes, so the pipeline emits an embedding queue that is the union of three levels, 34,072 units in total:

- **pericope level** (2,610): every pericope that was *not* split;
- **chunk level** (431): split pericopes enter only as chunks;
- **verse level** (31,031): every verse individually, for fine-grained exact retrieval (printed verse ranges such as “1–2” are kept as single units).

This queue is an artifact designed *for retrieval*. Critically for what follows, the early entity-extraction pipeline consumed this same queue as its input feed. That single wiring decision — extraction feed = embedding queue — silently propagated

TABLE I
CORPUS AND CHUNKING STATISTICS (CUV, NEW PUNCTUATION)

Item	Value
Books	66
Chapters	1,189
Pericopes	2,779
– split into chunks	169 (6.1%)
Chunks (from split pericopes)	431
Corpus size (characters)	1,329,046
– inside split pericopes	243,773 (18.3%)
Pericope length, median / P90 / max (chars)	388 / 931 / 3,544
Split-pericope length, median / max (chars)	1,353 / 3,544
Chunk length, median / max (chars)	644 / 796
Chunker: split threshold / target / min (tokens)	768 / 512 / 128
Chunker: overlap	1 verse
Tokenizer	BGE-M3

an embedding-oriented granularity into the knowledge graph and became the structural root cause (M1) of the defect chain quantified in Section VIII.

IV. KNOWLEDGE GRAPH CONSTRUCTION

The knowledge graph comprises three subnetworks — the corpus hierarchy, the entity network, and the cross-reference network — built by the four pipelines described below. Throughout construction we enforce a single anti-hallucination discipline, *grounding*: LLMs never invent entities, relation types, or evidence; they only *classify* candidates that rule-based stages propose, and every LLM output must quote an evidence span that is verified post hoc to be a literal substring of the source text, otherwise the output is demoted or discarded. This is the schema-based end of the paradigm spectrum surveyed in [26], and the structured-prompting choice follows the noise-suppression evidence of [7], [8].

A. Grounded Dual-Track Entity Extraction

Entity extraction targets six types (Table II) and runs two tracks in parallel: a named-entity-recognition (NER) track and a grounded classification track.

TABLE II
ENTITY TYPES (LIVE GRAPH COUNTS, JULY 6, 2026)

Type	Examples	Count
Person	Abraham, Moses, Jesus	2,419
Object	Ark of the Covenant, tabernacle	2,200
Event	the Exodus, the Resurrection	1,714
Place	Jerusalem, Egypt	1,299
Theme	redemption, grace, faith	987
Group	Israelites, Pharisees	505
Total		9,124

Live counts differ slightly from the raw extraction output (9,120; Section IV-A) because of import-time merging and the post-audit curation of Sections IX–X (16 generic-noun events deleted, 18 curated events added, one entity retyped); the pre-remediation live count, used by the audit of Section VIII, was 9,122.

1) *NER Track (Person / Place / Group)*: CKIP Transformers [17] performs Chinese word segmentation and named entity recognition (BERT-base models), with the type mapping PERSON→Person, GPE/LOC→Place, ORG/NORP→Group. Results are merged with a hand-built biblical entity dictionary using dictionary-first, longest-match precedence. Mention positions are located by literal string matching (`text.find()`) — a design whose consequences (no coreference; substring false hits) are quantified in Section VIII.

2) *Grounded Track (Event / Object / Theme)*: Four phases, rule-first with LLM escalation:

- 1) **Pericope-title mining**. Titles are typed by suffix and keyword heuristics (altar/vessel/ark suffixes → Object; redemption/grace keywords → Theme; residual default → Event). A 24-word generic-title stoplist (“day(s)”, “firstborn”, ...) intercepts generic nouns before the Event default — a guard added after the noise audit of Section VIII.
- 2) **POS candidate mining**. CKIP part-of-speech tagging proposes nouns (Na/Nb/Nv/Nc) with corpus frequency ≥ 3 ; adjacent nouns are compounded (e.g., “burnt” + “offering”).
- 3) **Rule classifier**. Exact lexicons and verb-component features assign types; candidates with confidence ≥ 0.8 are finalized without any LLM call.
- 4) **LLM classifier (grounded)**. Remaining candidates are classified by a small local model (Gemma 3 4B [37]) under two hard constraints: it may only label *provided* candidates (never add new ones), and it must return an evidence quotation that post-hoc verification confirms is a substring of the grounding text; failures are demoted.

An `EntityNormalizer` merges canonical names and aliases within each type and assigns `entity_id = {type}:{pinyin}` (toneless `lazy_pinyin` — a homophone-collision risk noted in Section XII). The extraction output is 9,120 entities and 173,896 mentions; mentions carry the granularity of their source unit (pericope, chunk, or verse), inherited from the embedding queue (Section III-E). Known upstream limitations, all revisited in the audit: a stopword list removes 33 single-character words including “God”, “Spirit”, and “Lord”; there is no coreference resolution; entity de-

TABLE III
CLOSED RELATION ONTOLOGY (37 TYPES)

Type pair (count)	Relations
Person×Person (13)	FATHER_OF, MOTHER_OF, SON_OF, DAUGHTER_OF, SIBLING_OF, SPOUSE_OF, ANCESTOR_OF, DESCENDANT_OF, TEACHER_OF, DISCIPLE_OF, ENEMY_OF, ALLY_OF, SUCCEEDED_BY
Person×Place (7)	BORN_IN, DIED_IN, RULED, VISITED, EXILED_TO, RETURNED_FROM, BUILT
Person×Object (4)	POSSESSED, RECEIVED, GAVE, DESTROYED
Person×Event (3)	PARTICIPATED_IN, INITIATED, VICTIM_OF
Person×Group (3)	MEMBER_OF, LEADER_OF, FOUNDED
Place×Place (2)	LOCATED_IN, NEAR
Event×Place (1)	OCCURRED_IN
Event×Event (2)	PRECEDED_BY, CAUSED
Group×Place (2)	ORIGINATED_FROM, SETTLED_IN

Each type defines domain/range, direction, inverse type, regex trigger signals, few-shot examples, and stage-specific confidence priors.

scriptions are 100-character text snippets; and pinyin IDs can collide across homophones.

B. Grounded Relation Extraction with a Closed 37-Type Ontology

Entity–entity fact edges are extracted by a five-stage pipeline against a closed ontology of 37 relation types (Table III), each declaring domain/range types, direction, inverse, textual trigger signals, few-shot examples, and per-stage confidence priors. Closing the schema prevents the LLM from inventing relation names, and typing the domain/range prunes the candidate space before any model call.

- 1) **Pair mining**. Entity pairs co-occurring in the same pericope are enumerated, kept only if some ontology type admits their type signature, and capped at 80 pairs per pericope.
- 2) **Rule classifier**. Per-type regex trigger signals fire when the two entities are within 25 characters; matches with confidence ≥ 0.85 are finalized.
- 3) **Domain priors**. 71 curated golden genealogy relations bypass the LLM entirely (64 of them enter the graph as distinct prior-stage edges; the remainder coincide with earlier-stage matches).
- 4) **Grounded LLM classification**. A local Gemma 4 31B model [38] — the relation stage was built after the entity stage and uses a larger model; the entity classifier’s 4B scale is itself flagged in the audit — must select exactly one type from the pair’s type-legal candidate set or answer NONE, under a JSON grammar constraint; the returned `evidence_span` must pass substring verification against the source pericope (doubly checked post hoc), otherwise the pair is rejected. This addresses the classic failure mode of triples divorced from their source context [32].
- 5) **Inverse materialization**. Symmetric/inverse types (FATHER_OF ↔ SON_OF) are materialized bidirectionally at confidence $\times 0.9$.

The full run (10–20 hours offline, checkpointed and resumable) yields 6,958 imported relations (772 rule, 64 prior, 5,370

LLM, 752 inverse) — and, importantly, 77,953 *unclassified* candidate pairs (LLM answered NONE) retained on disk with their source-pericope provenance. This “rejected pile” later became the raw material of the P0 event-layer rescue (Section IX). A companion description generator (Gemma 4 E4B) targets the $\sim 4,223$ Person/Place/Group entities with empty descriptions, generating ≤ 80 -character summaries grounded in the titles of mentioning pericopes; entities without pericope anchors cannot be described this way — which is why the audit of Section VIII still finds 46% of NER-type entities description-less.

Every fact edge carries confidence, evidence_span, source_pericope_id, and extraction_phase, so all graph facts are source-traceable and confidence-filterable — the property that later allowed co-occurrence backfill edges (confidence 0.35) to coexist distinguishably with LLM-extracted edges.

C. Cross-Reference Network

Cross-references (“chain references”) connect pericopes across books — prophecy and fulfillment, parallel gospel accounts, quotations. The network merges three sources of decreasing editorial cost:

- 1) **Printed parallel-passage annotations** parsed from the CUV Markdown: 774 edges (synoptic parallels and psalm titles);
- 2) **A curated list of famous New-Testament-to-Old-Testament (NT→OT) citations**: 142 edges typed as *quotation* or *allusion*;
- 3) **The Treasury of Scripture Knowledge (TSK)** [5], a public-domain cross-reference compendium with community vote counts: 344,799 raw rows are filtered (1,166 negative-vote rows, 9,811 self-loops) and mapped verse→pericope through a reverse index built from the embedding queue (covering 31,102 verses, with printed verse ranges expanded); only 7 rows fail to map, yielding 250,358 unique pericope pairs.

After idempotent merge the live graph holds **250,418** CROSS_REFERENCES edges. Every TSK edge stores its votes; manual edges are assigned a sentinel vote of 999 (highest trust). This vote-based provenance is what later made *selective de-noising* possible without deleting data (Section X) — the same “repair by re-weighting, not removal” philosophy advocated for KG denoising in [30].

D. Graph Schema and Storage

Figure 2 shows the schema; Table IV the live scale. Salient design points:

- **Dual labels.** Hierarchy nodes carry (:Pericope:Bible), entity nodes (:Person:Entity): the generic label supports whole-structure queries, the specific label exact matches.
- **No verse nodes.** Verse-level IDs (gen:1:0:v:3) exist only in Qdrant and PostgreSQL. This was a deliberate economy at design time — and the direct precondition of the silent mention-loss defect (Section VIII, M2).

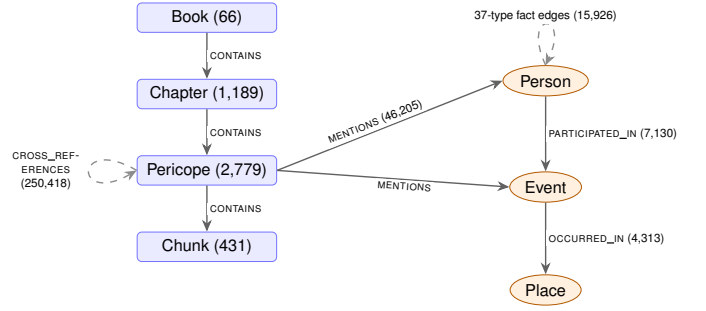


Fig. 2. Knowledge-graph schema: corpus hierarchy (left), entity network (right), the pericope-to-pericope cross-reference network (774 markdown + 142 curated + TSK edges), and sequential NEXT / NEXT_BOOK edges (omitted for clarity). Verse-level nodes deliberately do not exist — the design trade-off whose downstream cost is quantified in Section VIII.

TABLE IV
LIVE KNOWLEDGE-GRAPH SCALE (NEO4J, JULY 6, 2026)

Item	Count
Nodes, total	13,589
structure (Book/Chapter/Pericope/Chunk)	4,465
entities (six types)	9,124
Relationships, total	319,988
MENTIONS	46,205
CROSS_REFERENCES	250,418
semantic fact edges (37 types)	15,926
PARTICIPATED_IN / OCCURRED_IN	7,130 / 4,313
SON_OF / FATHER_OF / POSSESSED	659 / 647 / 551
structural: CONTAINS / NEXT / NEXT_BOOK	4,399 / 2,975 / 65

- **Entity properties.** canonical_name, aliases (a native list), description, mention_count.
- **Provenance-first edges.** Fact edges keep confidence and evidence; cross-reference edges keep votes and source; backfilled edges are flagged — all imports are idempotent MERGE operations, so the graph can be rebuilt or rolled back deterministically.

E. Vector Indexes and Database Import

Three vector families are produced:

- **Dense:** BGE-M3 [16], 1,024-dimensional, normalized, for all 34,072 queue units;
- **Sparse:** BM25 [14] ($k_1=1.5$, $b=0.75$) over CKIP-segmented [17] text, with the query-side IDF vocabulary exported for the online service;
- **Entity vectors:** each entity is embedded from the recipe “name (aliases). description. commonly appears in: pericope titles” truncated to 200 characters, upserted idempotently (UUID5 keys) into a dedicated collection (9,122 points).

Imports populate PostgreSQL (6 tables; the authoritative store), Qdrant (bible_embeddings, dense-only; bible_embeddings_hybrid with named dense+sparse vectors; bible_entities), and Neo4j (structure, mentions, fact edges, cross-references). The import layer is where the audit later found the single most damaging defect of the whole pipeline — verse-level mentions silently dropped because their target nodes do not exist (Section VIII) — and where honesty counters were subsequently installed.

V. ONLINE RETRIEVAL ARCHITECTURE

The online service is a FastAPI application exposing `POST /api/v1/query`. Figure 3 shows the query pipeline; this section walks through it stage by stage.

A. Triple-Database Roles and Content Hydration

The three stores play strictly complementary roles:

- **PostgreSQL** is the authoritative source of truth: full text, hierarchy, entity mentions. Every retrieval strategy returns only IDs; final passage text is always hydrated by a single PostgreSQL lookup (`get_content_by_id()`), which resolves 3-segment IDs to pericopes, 4-segment IDs to chunks, and verse IDs to their parent pericope. This guarantees that no matter which index proposed a candidate, the generator always sees identical, authoritative text.
- **Qdrant** is the semantic entry point: dense, hybrid, and entity-vector search.
- **Neo4j** is the relational entry point: hierarchy traversal, entity networks, and cross-reference expansion.

Strategy weights are *priors* on the reliability of the proposing strategy, not similarity scores; deduplication keeps the maximum weight per ID and weights are only ever upgraded. Every retriever is wrapped in per-strategy error isolation: a failing strategy is recorded in a `strategy_errors` field of the response and never aborts the query.

B. Query Pipeline

A query passes through seven steps: (1) verse-reference detection (regex); (2) LLM intent classification returning `{intent, entities, keywords}`; (3) signal detection and route selection; (4) parallel multi-strategy retrieval for the selected route; (5) deduplication, whole-pool reranking, rank fusion, and pins; (6) context assembly from the top-5 passages; (7) constrained answer generation. Three request-level switches matter for evaluation: `semantic_only` bypasses steps 1–3 (the ablation baseline); `use_graph` gates all Neo4j strategies per request (A/B without container restarts); `retrieval_only` skips step 7 (the fast evaluation loop of Section VI-C). Responses expose observability fields — per-source strategy, raw and fused scores, route used, strategies used, and errors — which later made question-by-question forensics possible without re-instrumentation.

C. Intent Classification and Signal Detection

- **Verse parser**: regex over canonical and abbreviated Chinese book names (longest-match, length-descending) resolves “Romans 3:23”-style and “Genesis chapter 1”-style references.
- **Intent classifier**: the generation LLM returns JSON with `intent ∈ {verse_lookup, topic, person, event, cross_reference}` plus entity names and keywords; a detected verse reference forces `verse_lookup`.
- **Dictionaries**: person/place gazetteers are *reused from the build-side extraction dictionary*, matched by longest substring; event detection uses a curated list of ~54

Algorithm 1 Signal-driven route selection

Require: signals S (six booleans), intent i

```

1: if  $S$ .book_chapter_verse then
2:   return R1 (exact verse)
3: else if  $S$ .book_chapter  $\wedge$   $S$ .multi_book then
4:   return R5 (cross-book)
5: else if  $S$ .book_chapter then
6:   return R2 (chapter + semantic)
7: else if  $S$ .multi_book  $\vee i = \text{cross\_reference}$  then
8:   return R5 (cross-book)
9: else if  $S$ .multi_person then
10:  return R3 (person graph)
11: else if  $S$ .event_keyword then
12:  return R4 (event graph)
13: else if  $S$ .has_place then
14:  return R6 (place graph)
15: else
16:  return Fallback (semantic + book anchor)
17: end if

```

event keywords (Sermon on the Mount, division of the kingdom, Paul’s conversion, ...); book-name matches require ≥ 2 characters; LLM-returned entities are re-filtered through the same dictionaries.

Six boolean signals result: *has-book-chapter-verse*, *has-book-chapter*, *multi-book*, *multi-person*, *event-keyword*, *has-place*. There is deliberately *no query rewriting*: the raw user query reaches both the embedder and the reranker verbatim, so the lexical gap between modern phrasing and the century-old CUV vocabulary must be bridged by graph anchors and the fusion layer — a recurring causal thread in Sections IX–X.

D. Six-Route Decision Tree

Different question shapes need different evidence: “What does John 3:16 say?” should be answered by SQL lookup; “the relationship between Moses and Jethro” by person-graph traversal; “how Jeremiah’s new-covenant prophecy is applied in Hebrews” by cross-book expansion. A priority-ordered decision tree (Algorithm 1) maps the six signals to six routes plus a fallback; each route launches its strategies in parallel with the priors listed in Table V.

E. Retrieval Strategies

Eleven strategies — three of them the typed graph traversals — share the ID-only/hydrate-later contract of Section V-A:

- **semantic** — BGE-M3 query embedding against the passage collection, top-20; when hybrid search is enabled, dense and BM25-sparse branches are prefetched (50 each) and merged by RRF [15] inside Qdrant, degrading automatically to dense-only if the sparse vocabulary is unavailable.
- **verse_direct** — exact SQL lookup of a verse, verse range, or whole chapter.
- **sql_chapter / sql_supplement** — direct chapter fetch; and same-chapter completion for chapters already hit by other strategies.

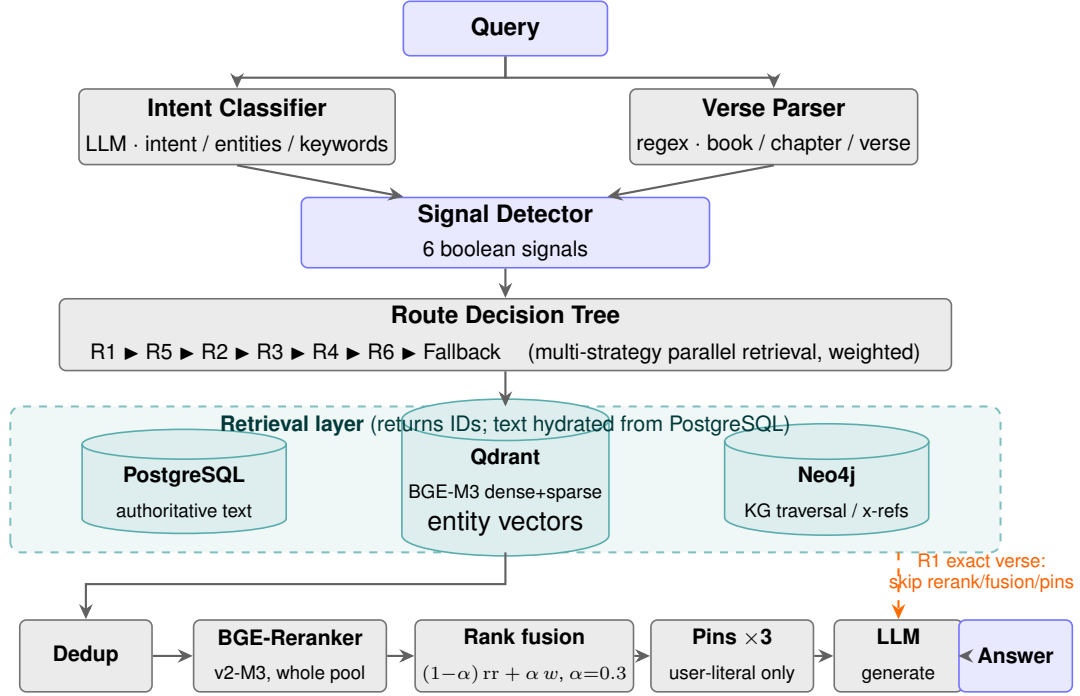


Fig. 3. Online query pipeline. Compared with the May 2026 deployment, the post-retrieval chain gained the rank-fusion stage, and the pin set was re-audited: only “user-literal” pins survive (Section V-F).

TABLE V
ROUTES AND STRATEGY COMPOSITIONS (WEIGHTS IN PARENTHESES)

Route	Strategies (strategy prior w)
R1	verse_direct (1.0); empty $\Rightarrow$ fall back to R2
R2	sql_chapter (0.9) + semantic (0.6)
R3	graph_person (0.9) + semantic (0.7) + entity_path + cross_ref_expand + entity_query (0.6) + sql (0.5) + book_anchor
R4	graph_event (0.85) + semantic (0.7) + cross_ref_expand + entity_query (0.6) + sql (0.5) + book_anchor
R5	cross_reference (0.85) sql_chapter (0.85) graph (0.75) + semantic (0.65) + entity_query (0.6) + sql (0.4) + book_anchor
R6	graph_place (0.85) + semantic (0.7) + entity_path + cross_ref_expand + entity_query (0.6) + sql (0.5) + book_anchor
FB	semantic/hybrid + book_anchor (if a book is named)

Names are the deployment’s route-configuration keys: graph denotes the route’s typed entity-graph traversal (graph_person / graph_event / graph_place); cross_reference is R5’s dedicated cross-reference retrieval, sharing the seed-support+votes machinery of cross_ref_expand; sql is a low-prior supplementary SQL lookup distinct from sql_chapter; hybrid is the dense+sparse mode of semantic. With use_graph=false, all Neo4j-backed strategies are gated off. Observed route distribution on the 100-question benchmark: R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2.

- **graph_person / graph_event / graph_place** — entity lookup by canonical name or alias (substring match, ordered by `mention_count`), then MENTIONS traversal to pericopes. Multi-person queries additionally retrieve *co-occurrence* pericopes at weight 0.9. Event anchors are ordered canonically (book/chapter ascending, so the narrative origin surfaces first) and tagged with `keyword_exact/anchor_rank` flags for the pin stage. All graph queries remap chunk IDs to their parent pericope *inside Cypher* — a fix that also restored retrieval visibility for 164 chunk-only entities (Section X).
- **cross_ref_expand** — ≤ 2 -hop expansion over CROSS_REFERENCES from seed pericopes, ranked by *seed support* (number of distinct seeds reaching the candidate) then by *votes*; 2-hop only fills unmet quota. Hop weights are provenance-stratified: manual edges 0.75/0.55 (1-/2-hop), TSK edges 0.60/0.50 — deliberately *below* the semantic prior 0.7 after the de-noising episode of Section X; expansion is capped at 10 candidates and seeds are chosen round-robin across strategies to prevent single-strategy monopoly.
- **entity_path** — multi-hop reasoning (≤ 2 hops) over the 37-type fact edges only (structural edges excluded), then MENTIONS to pericopes.
- **entity_query (EQ)** — the query embedding matched against *entity* vectors (top-8, cosine ≥ 0.4), then MENTIONS to pericopes, with hub-aware throttling: entities with `mention_count` > 50 contribute at most 3 pericopes, others 5, supplement cap 5. EQ is the designed bridge between modern question vocabulary and the classical corpus; its disable–re-enable history is itself an experimental finding (Sections VII and X).
- **book_anchor** — book-filtered semantic search guaranteeing that an explicitly named book contributes seeds.

TABLE VI
PIN LADDER (ADDITIVE BONUSES ON THE FINAL SCORE)

Pin	Bonus	Status	Trigger
chapter	+0.010	active	user names “Book, chapter N ” ^a
EQ	+0.005	legacy only	high-confidence EQ candidate
book-anchor	+0.004	active	user names a book
graph	+0.003	legacy only	reranker top-1 < 0.3
uncertainty			
keyword-exact	+0.002	active (fusion)	dictionary event keyword = entity name/alias ^b
event			

^a Guarantees ≥ 2 passages of the named chapter survive (candidates with $w \geq 0.85$ qualify).

^b Exact string equality only; hub entities (`mention_count` > 25) excluded; one anchor per event, at most two pins. The bonuses are spaced so pins never override one another.

F. Rank-Fusion Layer

The post-retrieval chain is the architectural core of the current system and the layer where two of our three incident rounds were resolved:

- 1) **Deduplication:** same-ID candidates keep the maximum strategy prior.
- 2) **Whole-pool reranking:** BGE-reranker-v2-m3 (the cross-encoder companion of [16]) scores every (query, passage) pair; scores are sigmoid-normalized to $[0, 1]$. The *whole* candidate pool is scored — not a top- k prefix — so fusion can promote candidates the reranker puts arbitrarily deep.
- 3) **Linear rank fusion:** $s_{\text{fused}} = (1 - \alpha) s_{\text{rr}} + \alpha w$ with $\alpha = 0.3$, where s_{rr} is the normalized reranker score and w the strategy prior of Table V. α is overridable per request (enabling the sweep of Section X-C); $\alpha = 0$ recovers pure reranker ordering, and a configuration flag restores the full legacy path.
- 4) **Pins:** a small additive ladder applied after fusion, restricted to “user-literal” triggers (Table VI).

Why fusion is necessary. The cross-encoder is a surface-form matcher. When modern question vocabulary does not occur in the CUV text, the ground-truth anchor’s reranker score can be as low as 0.001 — e.g., “the Last Supper” against a corpus that says “Passover meal” — and no amount of candidate-pool engineering makes it visible under pure reranker ordering. Fusion lets a candidate with $s_{\text{rr}} \approx 0.001$ but a graph prior of $w = 0.85$ enter the top-5. The magnitudes are interpretable: at $\alpha = 0.3$ a graph anchor ($w \in [0.85, 0.9]$) beats a TSK neighbor ($w \in [0.5, 0.6]$) unless the latter’s reranker advantage exceeds ≈ 0.11 . R1 (exact verse) bypasses reranking, fusion, and pins entirely; if the reranker itself fails, ordering falls back to strategy priors.

Pin lifecycle. The EQ pin and the graph-uncertainty pin were designed *before* fusion existed, as discrete rescues for graph evidence that pure reranker ordering was suppressing. Once fusion exposed those same signals continuously, the two pins were re-audited, found to be net-negative (they promoted $s_{\text{rr}} \approx 0.002$ noise to rank 1 on two benchmark questions), and retired from the fusion path — a mechanism-level lesson analyzed in Section XI. The surviving pins all encode “the

user literally named X” constraints, which are orthogonal to score fusion.

G. Answer Generation

The top-5 hydrated passages are formatted as numbered context blocks (“[N] Book, Chapter — pericope title (verse span)”), and the generator operates under a strict system prompt: answer *only* from the provided passages, cite book and chapter for every claim, add no outside knowledge or inference, respond in Traditional Chinese. The LLM provider is pluggable behind a factory (Ollama, Claude, OpenAI, Gemini); the deployed default is a locally served Gemma 4 E4B (8-bit quantized) at temperature 0.1 [38]. If retrieval returns nothing, the system refuses rather than answers. Responses are non-streaming JSON envelopes carrying the answer, per-source metadata (strategy, raw and fused scores), the route used, and timing — the observability that later enabled per-question forensics.

H. Deployment

The system ships as five Docker Compose services — backend (FastAPI), PostgreSQL 15 with pgvector, Qdrant v1.13.2, Neo4j 5.15 Community with APOC, and Ollama with GPU access — with health-checked startup ordering. The backend image is baked (no code volume); the only code-relevant mounts are the read-only BM25 vocabulary and the HuggingFace model cache. Build (`scripts/`), serving (`backend/`), and evaluation (`evaluation/`) each pin their own dependency manifest, isolating environment drift. All models — embedder, reranker, extractor, generator, judge — run locally; no external API is required for any pipeline stage, which is the cost/privacy design goal carried over from the system’s inception.

VI. EVALUATION FRAMEWORK

A. Benchmark

The benchmark contains 100 Traditional Chinese questions, 20 in each of five categories, with ground truth approved by annotators with biblical expertise:

- **VERSE_LOOKUP** — exact reference lookup (“What does Romans 3:23 say?”);
- **TOPIC** — thematic aggregation (“What does the Bible teach about forgiveness?”);
- **PERSON** — character-centric questions, often multi-person (“What was the relationship between Elijah and Elisha?”);
- **EVENT** — event-narrative questions (“What happened at the Last Supper?”);
- **GENERAL** — open questions requiring knowledge integration, including cross-book prophecy-fulfillment.

Each item carries `reference` (gold pericope-level provenance), `expected_answer_points` (a checklist of content points), and `reference_answer` (a model answer). The taxonomy deliberately spans exact matching, semantic search, entity-centric retrieval, and cross-book integration, so that route-level behavior is observable per category.

B. Metrics

Thirteen metrics fall into four families:

- 1) **Deterministic retrieval metrics** (no LLM, reproducible bit-exactly): hit rate, recall@5, precision@5, F1@5, MRR, MAP@5, NDCG@5, computed programmatically against the gold references at pericope granularity.
- 2) **RAGAS generation metrics** [35]: faithfulness, answer relevancy, context recall, and answer correctness, judged by a locally served Gemma 4 26B.
- 3) **Answer coverage** (ours): an LLM judge scores each expected answer point as covered (1.0), partial (0.5), or missing (0.0); coverage is the mean. Unlike F1-style correctness it is a *pure recall* metric — it rewards completeness without penalizing extra detail, complementing faithfulness as a hallucination check.
- 4) **Semantic similarity**: embedding cosine between generated and reference answers.

Three measurement pitfalls, encountered concretely in this project, qualify every LLM-judged number we report. (i) Per-question judge noise for RAGAS metrics reaches ± 0.4 , so 20-question per-category deltas below ≈ 0.05 are treated as noise. (ii) RAGAS answer relevancy applies a “noncommittal” rule that zeroes refusals; an honest “the provided passages do not state this” is scored 0, which we found systematically deflates cautious models (a commercial model’s relevancy rose above the local model’s once refusal-affected items were excluded). (iii) Position and verbosity bias make naive LLM-judged A/B comparisons overstate differences [11]. Retrieval conclusions in this paper therefore rest on the deterministic family wherever possible.

C. A/B Infrastructure and Reproducibility

Three properties of the harness matter for the experiments that follow:

- **Per-request configuration**: `use_graph` and `fusion_alpha` are request-level overrides, so graph/semantic A/B tests and α sweeps run against one container without restarts or rebuilds.
- **Fast retrieval loop**: with `retrieval_only=true` the backend skips generation; the 100-question retrieval suite completes in ~ 15 minutes versus ~ 2.5 hours for the full pipeline (36 min collection + 1 h 42 m RAGAS + 13 min coverage), with retrieval metrics verified *bit-exact* against the full pipeline.
- **Re-scorable raw outputs**: every run archives raw responses (sources, strategies, scores per question); metrics can be recomputed from archives, and a comparison tool emits per-question deltas and hit-flip lists — the instrument behind every forensic table in Sections IX and X.

VII. ROUND 0: SIGNAL-ADAPTIVE RETRIEVAL VS. SEMANTIC-ONLY

The first experiment round (May 2026) established the value of the architecture itself, before any knowledge-graph optimization.

TABLE VII
ROUND 0 — SIGNAL-ADAPTIVE VS. SEMANTIC-ONLY (100 QUESTIONS, MAY 2026)

Category	Metric	Signal-adaptive		Semantic-only	
		Claude	Gemma	Claude	Gemma
Retrieval	Hit rate	0.960	0.950	0.810	0.810
	Recall@5	0.886	0.889	0.758	0.758
	MRR	0.805	0.802	0.647	0.647
Generation	Faithfulness	0.957	0.951	0.899	0.903
	Relevancy	0.815	0.893	0.709	0.774
	Context recall	0.797	0.791	0.680	0.677
	Coverage	0.750	0.722	0.618	0.628

Retrieval metrics under semantic-only are identical across answer models by construction (deterministic retrieval); small graph-mode differences arise because the answer LLM also performs intent classification, which shapes graph-traversal inputs on R3–R6.

TABLE VIII
ROUND 0 — ANSWER COVERAGE BY QUESTION TYPE (SIGNAL-ADAPTIVE)

Type	Claude	Gemma	Δ
VERSE_LOOKUP	1.000	0.975	−0.025
TOPIC	0.900	0.856	−0.044
PERSON	0.635	0.594	−0.041
EVENT	0.613	0.606	−0.007
GENERAL	0.601	0.580	−0.021

A. Setup

Four configurations cross two retrieval modes with two answer models on identical code: {signal-adaptive (graph), semantic-only} $\times$ {Claude Haiku 4.5 [39] (commercial API), Gemma 4 E4B [38] (local, zero cost)}, with temperature 0.1 and top-5 context. The semantic-only baseline bypasses signal routing and all graph strategies, using pure dense retrieval. At this time point the entity-query strategy was *disabled*: an earlier deployment had found that cross-encoder reranking systematically displaced its candidates — surface-form scores beat entity anchors — so EQ was turned off pending a mechanism fix. This early incident is the first of the three rounds of ranking-layer evidence assembled in Section XI.

B. Results

Three findings (Tables VII and VIII):

Signal-adaptive retrieval is the dominant quality lever.

The architecture beats the semantic-only baseline on every metric, with decisive retrieval gains: hit rate +15 pp (0.81 $\rightarrow$ 0.96), recall@5 +13 pp, MRR +16 pp. The improvement propagates to generation (faithfulness 0.90 $\rightarrow$ 0.95, coverage 0.62 $\rightarrow$ 0.74, two-model averages) and is *model-independent*: Claude and Gemma improve by near-identical margins, isolating retrieval architecture as the causal factor.

A local model matches the commercial API. Holding the pipeline fixed, the two models differ by under 4 pp on six of seven metrics; only answer relevancy differs more (7.8 pp, in Gemma’s favor — partly the noncommittal-refusal artifact of Section VI-B). Strengths are complementary (Claude leads faithfulness and coverage, Gemma relevancy); neither dominates. The practical consequence: the system runs at zero

marginal cost with full data privacy, and the investment frontier is retrieval infrastructure, not model scale.

The hard question types point at the knowledge graph. VERSE (1.000/0.975 coverage) and TOPIC (0.900/0.856) are largely solved by exact SQL and semantic search; PERSON (0.635/0.594), EVENT (0.613/0.606), and GENERAL (0.601/0.580) lag. Person questions hinge on entity mentions and their (absent) coreference; multi-chapter narratives strain fixed-window retrieval; and at this time point the cross-reference network had only 916 edges to serve cross-book questions. Notably, PERSON was the *only* category where graph mode trailed the semantic baseline (hit 0.95 vs. 1.00) — a deficit that the audit traces to entity-layer damage rather than to routing. These observations motivated the audit of Section VIII.

VIII. KNOWLEDGE-GRAPH QUALITY AUDIT: THE GRANULARITY-MISMATCH CAUSAL CHAIN

Guided by the hypothesis that “KG quality problems trace back to the original chunking design,” we audited the entire construction pipeline on July 5, 2026, with every number re-verified live (Cypher queries, JSONL recomputation, line-level code reading). The hypothesis was confirmed — *but only after being made precise*: the damage was caused not by chunking parameters (512/768 tokens, 1-verse overlap are all defensible) but by the **silent inheritance of an embedding-oriented granularity structure by three downstream layers** (extraction, import, retrieval).

A. Eight Quantified Gaps

Table IX lists the eight gaps with their live evidence. The three heaviest (#2, #3, and the split-pericope blindness described below) are all granularity-inheritance effects; the others are missing pipeline features (no coreference, no entity resolution, no alias generation) that chunk boundaries then amplify.

B. Six Mechanisms (M1–M6)

M1 — Extraction feed = embedding queue (structural root cause). The NER track consumed `embedding_queue.jsonl` directly: 2,610 pericopes + 431 chunks + 31,031 verses. This queue’s granularity was designed for retrieval (split pericopes appear only as chunks; every verse appears individually); extraction inherited it unconditionally, so most NER mentions were emitted at verse granularity.

M2 — Import-layer granularity mismatch: 55.9% of mentions evaporate. Verse-level mentions have no corresponding Neo4j node (Section IV-D); the import MATCH silently no-ops and 97,235 mentions vanish while the log reports success. Recomputing offline showed the loss was fully recoverable: 100% of verse source IDs can be remapped to their parent pericope by stripping the verse suffix, which would add 5,853 unique (entity, pericope) anchor pairs and re-anchor 1,310 of the 1,474 blind entities (89%).

M3 — Two-layer misalignment: 18.3% of the text doubly blind. For the 169 split pericopes, NER-type entities (Person/Place/Group) anchor only at chunk/verse level, while *both* the graph retrievers and the relation-extraction pair miner match only pericope-level MENTIONS. Entities of these Old Testament narrative passages were therefore invisible to retrieval *and* excluded from relation extraction — precisely the $\text{person} \times \text{event}$ material of the hardest benchmark categories.

M4 — Chunk boundaries $\times$ absent coreference. The pipeline has no coreference resolution anywhere; mentions are located by literal string search, and the 1-verse overlap rarely contains the antecedent of “his father-in-law” or “he.” Document-level relation extraction quantifies the ceiling this imposes: at least 40.7% of relational facts require cross-sentence evidence and 17.6% require coreference reasoning [6]. (We found no peer-reviewed quantification of overlap-size effects on extraction, and therefore rely on this indirect evidence.) The kinship edges missing for Jethro (Exodus 18) — where the text says “Moses’ father-in-law” with the name far away — are the local instance.

M5 — Mixed granularity: duplicated and inconsistent extraction. The same text is extracted at up to three granularities; the same entity may be found at fine granularity and missed at coarse, with only toneless-pinyin surface forms as deduplication keys (which also *over-merge* homophones). Extraction granularity directly controls density — GraphRAG’s ablation found nearly twice the entity references at 600-token versus 2,400-token chunks [3] — so mixing granularities guarantees inconsistency; and atomic-fact decomposition work confirms finer units extract more completely [34]. Cross-chunk relation loss is a known systematic failure of chunk-local extraction [33], and retrieval-augmented construction [31] exists precisely to re-assemble such context.

M6 — 512/768 tokens was never an extraction constraint. The 169 split pericopes have median length 1,353 characters (max 3,544) — far below the context degradation range documented by [27]. Extraction could simply have consumed whole pericopes; chunks should have remained a storage/retrieval concern. (The “storage unit $\neq$ extraction unit” separation is standard in production KG builders, e.g., Neo4j’s LLM Graph Builder recombines chunks before extraction [40].)

C. Attribution and the Cross-System Survey

The audit’s precise conclusion: *the chunking parameters are innocent; the unconditional inheritance of chunking granularity by extraction, import, and retrieval is guilty*. Independent of granularity, the missing-feature defects (no coreference, no entity resolution, no alias generation, snippet descriptions) have direct literature counterparts: type-aware coreference cuts node duplication by 33.28% [7]; removing coreference alone raises duplication by 28.25% while removing structured prompting mainly raises noise [8]; three-stage LLM coreference achieves -45.21% duplication with the benefit *growing* with document length [9] — longest for exactly the split pericopes our pipeline handles worst.

We also surveyed nine production GraphRAG codebases (Microsoft GraphRAG, LightRAG, nano-graphrag, fast-

TABLE IX
THE EIGHT AUDITED GAPS (JULY 5, 2026, PRE-REMEDIATION; ALL NUMBERS VERIFIED LIVE)

#	Gap	Live evidence	Mechanism
1	Empty aliases	35 of 9,122 entities (0.38%) have any alias; dictionary-graph naming mismatches make alias fallback dead code	pipeline gap
2	Silent mention loss	97,235 of 173,896 mentions (55.9%) are verse-level; the importer’s MATCH finds no verse node, the MERGE silently no-ops, and the counter still reports “173,896 imported”	M1+M2
3	Retrieval-blind entities	1,474 entities (16%) have no pericope anchor (1,030 zero-anchor + 444 chunk-only); graph retrievers match only (Pericope) – [:MENTIONS]	M2, M3
4	Duplicate-name splits	878 names split across 1,900 nodes, all cross-type (e.g., the Ark appears as Group/Object/Person/Place); alias-based merging is inert because aliases are empty	pipeline gap × M5
5	Hollow event layer	96.2% of 1,712 events have no temporal/causal edge (45 edges by the audit’s count); 67.9% lack a place and 65.6% lack any participant (construction-CI measurement)	pipeline gap × M3
6	Relation import rate 8.1%	85,439 candidate pairs → 6,958 imported; 77,953 unclassified retained with provenance	pipeline gap
7	Noise entities	the conjunction “dan” (but) indexed as the place Dan with hundreds of anchoring sources; “Jehovah” typed as Group; generic “day(s)” event with 413 mentions	pipeline gap
8	Upstream extraction quality	4B-parameter classifier; 33 single-character stopwords remove God/Spirit/Lord; descriptions are 100-char snippets (46% of NER-type entities have none), polluting entity vectors — an upstream cofactor of the May EQ displacement	pipeline gap

graphrag, Neo4j LLM Graph Builder, LazyGraphRAG, KAG, iText2KG, RAPTOR) at source level. Four regularities recur: (a) index-time deduplication is exact-name only, with approximate entity resolution deferred to post-hoc/incremental stages keyed on lexical-distance + embedding-similarity double signals; (b) merged surface forms are written back as aliases — which is *the* correct alias-generation mechanism our pipeline lacks; (c) multi-round gleaning re-extraction exists only in the GraphRAG lineage [3] (the paper describes forced-continuation gleanings; cost-sensitive systems omit it); (d) mainstream systems extract at a *single* unified granularity — our three-granularity mix with no resolution stitching is the outlier and the fragmentation driver.

IX. P0 REMEDIATION AND THE NEGATIVE RESULT

A. Six Data Repairs (No Re-Extraction)

P0 repaired data in place — days of work, no re-extraction — following the audit’s cheapest-first ordering (Table X). Construction-side quality indicators moved decisively; three of them are now tracked as permanent CI metrics.

B. Evaluation Conditions

The 100-question graph-mode evaluation was re-run under conditions verified identical to the May baseline: same answer model (Gemma 4 E4B), same judge (Gemma 4 26B), same disabled entity query, and — verified question by question — the *same route distribution* (R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2). The only systematic difference was the state of the knowledge graph. Retrieval metrics are deterministic, so every delta below is exactly reproducible from archived raw responses.

C. Result: Construction Metrics Up, Retrieval Metrics Flat

Table XI is the paper’s central negative result: after a completeness overhaul that raised anchor coverage by 14.4 points and multiplied cross-references by 273, **every deterministic retrieval metric was flat or slightly down**, and answer-side

TABLE X
P0 DATA REPAIRS AND CONSTRUCTION-SIDE RESULTS (JULY 6, 2026)

Repair	Result
Verse-mention backfill	+5,853 pericope anchors; blind entities 1,474 → 164; anchor coverage 83.8% → 98.2% (prediction from the audit’s offline re-computation matched exactly)
Import honesty	importer rewritten: verse remap + honest counters (silent-drop class eliminated for future rebuilds)
Alias injection	38 nodes gain dictionary aliases (the dictionary’s coverage ceiling; mass aliases require entity resolution)
Noise gating	“dan” anchors 759 → 26 after classifying 1,882 mentions (44 geographic, 1,439 conjunction, 399 substring false hits inside Satan/Nathanael/Dedan/Abidan/Medan); 16 generic-noun events deleted across all three stores; Jehovah retyped Group→Person
Unclassified-relation rescue	Event-related pairs reclassified from the 77,953-pair pile: +5,641 PARTICIPATED_IN, +3,419 OCCURRED_IN (confidence 0.35, flagged as co-occurrence backfill); events with participants 34.4% → 84.0%, with places 31.8% → 75.5%
TSK import	CROSS_REFERENCES 916 → 250,418 (×273); retrieval-side expansion simultaneously switched to seed-support+votes ordering

TABLE XI
P0 NEGATIVE RESULT — OVERALL METRICS (100 QUESTIONS, GRAPH MODE)

Metric	Pre-P0 (May 16)	Post-P0	Δ
Hit rate	0.9500	0.9300	−0.0200
Recall@5	0.8892	0.8658	−0.0234
NDCG@5	0.8563	0.8496	−0.0067
MRR	0.8020	0.7978	−0.0042
Context recall (RAGAS)	0.7908	0.7767	−0.0141
Answer coverage	0.7223	0.7253	+0.0030
Answer correctness (RAGAS)	0.5874	0.5960	+0.0086
Faithfulness (RAGAS)	0.9513	0.9564	+0.0051

movements were inside judge noise. By category: VERSE

and TOPIC unchanged at 1.00 hit rate; PERSON unchanged *question by question* (still 0.95, still losing to the semantic baseline’s 1.00); EVENT dropped from 0.90 to 0.80 hit rate and its context recall from 0.608 to 0.546; GENERAL recall slipped from 0.654 to 0.638. The two indicators the audit had designated as success criteria — PERSON hit rate and EVENT context recall — both failed to move in the predicted direction.

D. Mechanism Diagnosis (Question-by-Question)

Every regressed or unmoved question was traced through its archived sources. Three mechanisms, each verified rather than conjectured:

1) *TSK votes are a double-edged sword*: After TSK import, cross-reference expansion went from “most questions have no neighbors” to “any seed reaches ~ 180 one-hop neighbors,” and the expansion ranked candidates by community votes. But *high votes mean strong topical-theological association, not same-event narrative*. Table XII shows the five forensic cases: in each, the displacing passages are topically related by the cross-reference logic yet wrong for the narrative question. The same mechanism *helped* topical aggregation questions (TOPIC coverage $+0.031$; the Sermon on the Mount correctly gained its Sermon on the Plain parallel), so the repair had to be category-sensitive re-weighting — not rollback.

2) *The reranker monopolizes the final ranking*: The final ordering at this time point was literally `sorted(passages, key=rerank_score)`: strategy priors and votes decided only *who entered the pre-rerank pool*, never the final rank. Consequently (a) construction improvements — correct anchors newly present in the pool — could not surface unless they also won the surface-form contest, and (b) construction side effects — topical noise whose surface forms match the query (“Paul,” “resurrection,” “prayer”) — displaced correct answers directly. This is the same bottleneck that had forced EQ off in May (Section VII); the P0 evaluation is its second independent confirmation.

3) *The benchmark reads only the head*: The 1,310 entities re-anchored by the backfill are long-tail entities that had been extracted only at verse level; the 100 questions ask about head entities (Elijah, David, Moses) whose pericope anchors were already plentiful. **On this yardstick, construction-side quantity had already saturated** — the decisive premise for deferring further construction investment. This is the local, deployment-level face of the benchmark-dependence phenomena reported in [10], [11], [25].

4) *Persistent failures (pre- and post-P0) typed by root cause*: EVENT_008 (division of the kingdom) and EVENT_014 (the Last Supper) fail because the graph event nodes are named differently (“revolt of the northern tribes”; “Passover meal”), have no aliases, and EQ — the designed lexical bridge — was off. PERSON_004 (the roles of Moses, Aaron, and Miriam) fails structurally: the answer is scattered across chapters and co-occurrence density favors genealogies/rosters; no top-5 pericope set suffices. That question is typed to the aggregation layer (community summaries / RAPTOR trees [3], [22]), not to P0/P1.

E. Decision: Defer Re-Extraction, Fix the Ranking Layer

The audit’s original roadmap continued with P1 (decoupled re-extraction, coreference, gleanings, entity resolution — week-scale). The diagnosis, however, showed that (i) P1’s gains concentrate in the long tail, which this benchmark cannot read (mechanism 3); (ii) until the ranking layer transmits graph signals, further construction investment is absorbed by the reranker (mechanism 2); and (iii) the remaining blind-entity population had already shrunk to 164 (1.8%). P1 was therefore **deferred**, and three day-scale, evaluation-verifiable ranking-layer repairs were scheduled instead — with the explicit note that P1’s value proposition is unchanged *as graph asset*; what failed was the claim that its effect would be observable on this benchmark.

X. RANK-FUSION REMEDIATION

A. Three Repairs and Three Chained Fixes

Repair 1 — TSK de-noising by provenance-stratified weights. Expansion cap $30 \rightarrow 10$; TSK edges (votes < 999) declassified to hop weights $0.60/0.50$ — deliberately below the semantic prior 0.7 — while manual edges keep $0.75/0.55$; the R5 blanket weight uplift for expansion candidates was removed. Rationale: keep the data, demote its priority; category-sensitive by *provenance*, not by deletion.

Repair 2 — Head-event backfill and EQ re-enablement. Motivated by a coverage audit: 31 of the 54 dictionary event keywords matched *zero* Event node by name — the graph calls the Last Supper a “Passover meal.” Eleven existing Event nodes received question-phrasing aliases and 18 curated Event nodes (56 MENTIONS edges, every anchor verified live) were added, synchronized across all three stores; the entity-query strategy was then re-enabled. This is a hand-curated, head-only instance of the P1 alias/entity-resolution program — days, not weeks.

Repair 3 — Linear rank fusion. The final ranking became $s_{\text{fused}} = (1 - \alpha) s_{\text{rr}} + \alpha w$ (Section V-F), with α overridable per request for the ablation.

Full-suite re-evaluation then triggered three *chained* fixes, each from a per-question regression trace: (a) graph and EQ retrievers now remap chunk IDs to parent pericopes inside Cypher, fixing a duplicate-seat bug where `exo:29:0` and its chunk `exo:29:0:0` both occupied top-5 seats (and, as a side effect, restoring retrieval visibility to the 164 chunk-only entities); (b) the legacy EQ pin and graph-uncertainty pin were retired under fusion after they promoted $s_{\text{rr}} \approx 0.002$ noise to rank 1 on two GENERAL questions; (c) a keyword-exact event pin was added for the one failure class fusion cannot reach — Acts 9 narrates Paul’s conversion calling him exclusively *Saul*, so every surface signal fails (reranker 0.074, rank 7) and only the curated dictionary $\rightarrow$ entity bridge can pin the anchor.

B. Results

Table XIII and Fig. 4 give the headline: hit rate 0.97, recall@5 0.906, context recall 0.830, coverage 0.766; EVENT hit rate 1.00 with context recall 0.728 — not merely recovering the P0 regression but surpassing the pre-P0 state on every

TABLE XII
FORENSICS — FIVE QUESTIONS WHERE VOTE-RANKED TSK EXPANSION DISPLACED CORRECT NARRATIVE ANCHORS (P0 EVALUATION)

Question	Pre-P0 sources (correct)	Post-P0 sources (displaced by)	Cross-reference logic of the intruders
EVENT_019 Paul’s conversion	Acts 9 (Damascus road), Acts 22, Acts 26 (his own retellings)	Acts 13 ×2, Acts 28, Romans 1	Pauline mission-theme chains (pre-P0 this question’s expansion found nothing; post-P0 it always fires)
EVENT_017 Resurrection day	Luke 24 (Emmaus), Mark 16	Mark 9, Matthew 17 (Transfiguration)	“Son of Man risen from the dead” prophecy chains
EVENT_015 Gethsemane	Luke 22	Romans 8, Ephesians 6	“prayer” theme chains
PERSON_011 Elijah & Elisha	1 Kings 19 (the call)	Matthew 27	“see whether Elijah will come” (crucifixion bystanders)
GENERAL_013 1 Cor. 15 vs. Genesis	1 Cor. 15, Isaiah 49	Acts 2, Romans 6, Revelation 21	resurrection/death theme chains

TABLE XIII
METRIC EVOLUTION ACROSS THE THREE TIME POINTS (100 QUESTIONS, GRAPH MODE, IDENTICAL DETERMINISTIC SCORING)

Metric	Pre-P0	Post-P0	Post-fusion	Δ^a
Hit rate	0.9500	0.9300	0.9700	+0.0400
Recall@5	0.8892	0.8658	0.9058	+0.0400
NDCG@5	0.8563	0.8496	0.8653	+0.0157
MRR	0.8020	0.7978	0.8335	+0.0357
Context recall	0.7908	0.7767	0.8297	+0.0530
Answer coverage	0.7223	0.7253	0.7664	+0.0411
Answer correctness	0.5874	0.5960	0.6173	+0.0213
Faithfulness	0.9513	0.9564	0.9495	−0.0069
EVENT hit rate	0.90	0.80	1.00	+0.20
EVENT context recall	0.608	0.546	0.728	+0.182
PERSON hit rate	0.95	0.95	0.95	0
PERSON context recall	0.667	0.683	0.745	+0.062
TOPIC / VERSE hit rate	1.00	1.00	1.00	0
Margin vs. semantic (hit)	+0.14	+0.12	+0.16	—

^a Post-fusion minus post-P0. Answer-side conditions identical to post-P0 (answer: Gemma 4 E4B; judge: Gemma 4 26B). Faithfulness flat across all three time points $\Rightarrow$ answer-side gains derive from better context, not generation drift. Semantic-only baseline: hit 0.81, recall@5 0.7575. LLM-judged rows report the documented run of record; re-judging the identical retrieval outputs with the same judge shifts them within the noise band of Section VI-B (e.g., faithfulness 0.9495 vs. 0.9655) — one more reason all retrieval conclusions rest on the deterministic rows, which are bit-exactly reproducible.

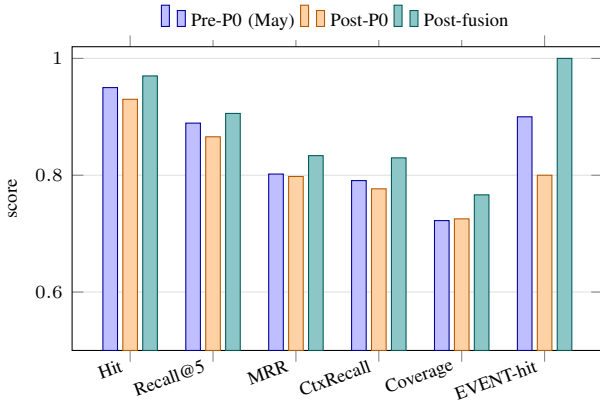


Fig. 4. The three-round shape of the evidence chain: the KG completeness overhaul alone (orange) reads flat-to-negative on a head-entity benchmark; unlocking the ranking layer (teal) recovers and surpasses every indicator.

TABLE XIV
 α ABLATION (RETRIEVAL-ONLY LOOP; ALL OTHER REPAIRS ACTIVE)

	$\alpha = 0$ (pure reranker)	$\alpha = 0.3$ (deployed)
Overall hit rate	0.97	0.97
EVENT hit rate	1.00	1.00
EVENT recall@5	baseline	+0.025
TOPIC MRR	baseline	+0.133
NDCG@5	baseline	−0.02

$\alpha = 0$ keeps the data repairs, chunk remap, pin changes, and keyword-exact pin; only the fusion term is disabled.

EVENT indicator; and the margin over the semantic baseline widened to +0.16 hit rate / +0.148 recall@5. Four questions flipped from miss to hit — the two persistent pre-P0 failures (EVENT_008 via aliases+EQ, EVENT_014 via curated node + fusion) and the two TSK casualties (EVENT_017 via TSK declassing, EVENT_019 via the keyword-exact pin) — and all five forensic cases of Table XII recovered, with EVENT_015 regaining full recall (its Matthew and Luke anchors returned to the top-5). The quick retrieval loop and the full pipeline agree bit-exactly on every retrieval metric.

Two residuals are instructive rather than problematic. GENERAL NDCG sits 0.08 below post-P0 — traced per question to the *retirement* of pin-based artificial top-1 placement (the correct passages remain in top-5; the earlier perfect MRR was pin manufactured, so the “drop” is the removal of an artifact, not a regression). PERSON_004 remains the single PERSON miss, by design out of scope for ranking repairs (Section IX-D).

C. α Ablation: A Complementary Structure

Sweeping the fusion coefficient with everything else fixed (Table XIV) decomposes the remediation:

- **Discrete repairs saturate hit rate.** At $\alpha = 0$, hit rate is already 0.97 and EVENT already 1.00: curated data (aliases, nodes) and the dictionary pin are responsible for *whether* the answer appears at all.
- **Continuous fusion recovers the tail.** $\alpha = 0.3$ adds EVENT recall@5 +0.025 — the multi-gospel parallel anchors of the Last Supper carry $s_{\text{rr}} \approx 0.01$ –0.02 and only fusion returns them — and TOPIC MRR +0.133,

TABLE XV
THREE INCIDENTS, ONE MECHANISM: GRAPH ASSETS GATED BY
RETRIEVAL-SIDE MACHINERY

Round	Asset in the graph	Gating mechanism	Resolution
May 2026	entity vectors (EQ) bridging modern phrasing	reranker surface-form monopoly	EQ disabled; re-enabled under fusion
P0 (July)	correct anchors newly in pool; TSK breadth	same monopoly + vote-ranked topical noise	fusion + provenance-stratified weights
Dual-anchor family	<i>all</i> needed cross-book edges present	1-hop expansion depends on seed correctness	open (query decomposition)

at the cost of NDCG@5 -0.02 (high-prior, low-surface-form candidates advancing).

- **The trade resolves by downstream use.** All top-5 passages enter the generation context, so recall dominates internal ordering; $\alpha = 0.3$ was adopted.

The two mechanisms substitute for the reranker’s surface-form monopoly at *different layers* — discrete repairs at the candidate/vocabulary layer, fusion at the scoring layer — and their effects are complementary rather than redundant. This decomposition is the constructive counterpart of the negative result: the same graph assets that read as zero in Section IX produce the Table XIII gains once the ranking layer transmits them.

XI. DISCUSSION

A. Three Co-Determinants of Observable Benefit

The central methodological claim of this paper is that in a GraphRAG system, **knowledge-graph construction quality, the retrieval ranking mechanism, and benchmark design jointly determine observable benefit** — and any one of the three can silently zero out investment in the others. The evidence chain is summarized by the paper’s three-round structure: a completeness overhaul whose construction metrics all moved (anchor coverage +14.4 pt, event completeness +50 pt, cross-references $\times 273$) read as *zero* through a reranker-monopolized ranking onto a head-saturated benchmark (Section IX); one day of ranking-layer repair then released more measured value than the week-scale re-extraction program was projected to deliver on the same benchmark (Section X). The practical rule we draw: *optimization order should follow the bottleneck layer, not the upstream-first order of the data flow* — and the bottleneck is found by per-question forensics, not by metric staring.

B. Three Independent Rounds of the Same Lesson

Table XV aligns the three incidents. The third deserves elaboration because it was discovered *after* the fusion remediation and is therefore an independent probe. The two remaining GENERAL misses are one family: cross-book *prophecy-fulfillment* questions requiring two anchor sets simultaneously (GENERAL_006: Jeremiah’s new covenant in Hebrews, gold = Jer. 31:31–34 + Heb. 8; GENERAL_008:

Zechariah’s prophecies in Passion Week, gold = Zech. 9:9; 11:12–13; 12:10 + Matt. 21/27). Live graph verification shows **every needed cross-reference edge exists** — Jer. 31 $\leftrightarrow$ Heb. 8 (3 edges), Zech. 9 $\leftrightarrow$ Matt. 21 (2), Zech. 11 $\leftrightarrow$ Matt. 27 (5), Zech. 12 $\leftrightarrow$ John 19 (2); TSK is *designed* for exactly this relation. The failure is upstream of ranking: retrieval seeds land in the wrong chapters (“prophecy” surface forms hit the false-prophet conflict of Jer. 26/28/29; “Zechariah” hits the Zerubbabel visions of Zech. 4), and 1-hop expansion from wrong seeds cannot reach right edges. Assets in the graph; value gated by the seed-selection mechanism. These two questions also post the two lowest answer-correctness scores of the entire suite (≈ 0.13 , stable across re-judged runs).

C. Retrieval-to-Answer Attenuation

Post-fusion, retrieval improved sharply (context recall +0.053) while answer correctness moved +0.021 — at the edge of judge noise. A clean single-question case isolates the attenuation: EVENT_005 achieves perfect retrieval (context recall 1.0, recall@5 1.0) yet its judged answer scores only 0.4 coverage and 0.173 correctness on the run of record — and re-judging the identical outputs moves those to 0.70/0.353, an instability that is itself the judge noise of Section VI-B at work. Faithfulness is flat across all three time points (≈ 0.95), so the gains genuinely come from better context; the loss occurs in the small generation model (Gemma 4 E4B) and in the judged metrics themselves. The pipeline’s quality bottleneck has, for these questions, *moved past retrieval* — consistent with our framing that the observable-benefit chain must be examined end to end. Any answer-model A/B here must first neutralize the noncommittal-refusal artifact of Section VI-B, or the honest model loses by construction — the local instance of the evaluation-bias warnings in [11].

D. Benchmark Saturation: Evaluation Design as Prerequisite

At 97/100 (VERSE, TOPIC, EVENT at 1.00 hit rate), the benchmark is saturated in the head, and the three residual misses belong to two families that need *new architecture*, not better ranking: cross-chapter aggregation (PERSON_004; the community-summary/RAPTOR layer [3], [22]) and cross-book dual-anchor conjunction (GENERAL_006/008; query decomposition). Consequently, *every* remaining roadmap item — long-tail re-extraction (P1), event-temporal modeling (P2, [23], [24], [34]), hierarchical aggregation (P3) — would read as zero on this yardstick: P1’s tail improvements are unread by head questions (proved empirically by P0), P2’s target questions already retrieve perfectly (their loss is in the attenuation chain above), and P3’s target population is $n = 1$. The methodological consequence: **benchmark redesign is the prerequisite for all further optimization**, not a reporting afterthought. We specify the needed instruments in Section XIII; the general principle — that the benchmark itself gates which layer’s investment is visible — echoes the defect-sensitivity methodology of BRINK [10] and the task-dependence findings of [25]. A cautionary corollary from Section X: with $n = 2$ family sizes, tuning the system on the residual misses would be overfitting; the family must be *populated first*, then fixed.

E. Patch Lifecycles Under Architecture Evolution

The pin audit (Section V-F) generalizes: the EQ pin and graph-uncertainty pin were correct *for the architecture they were built in* — discrete rescues under a ranking that discarded graph signal. Once fusion transmitted that signal continuously, the same pins became net-negative (noise promotion), and the “regression” observed when retiring them (GENERAL NDCG -0.08) was itself an artifact of the artificial placements they used to manufacture. Remediation machinery must be re-audited whenever the layer it compensates for changes; yesterday’s fix is today’s bug. We suspect this lifecycle is common in deployed RAG systems, where discrete patches accrete around ranking layers precisely because ranking is the last mile.

F. Cost and Privacy

All conclusions above were reached with every model local — embedder, reranker, extraction classifiers, generator, and judge — on a single GPU host. Round 0 established that a locally served 8-bit E4B-class model matches a commercial API model end to end on this task within 4 pp on six of seven metrics (Table VII); nothing in the later rounds required revisiting that conclusion, because all optimization value was retrieval-side. For sensitive religious-domain deployments — where both the corpus and user questions may be confidential — this is the operative cost/privacy result: the investment frontier is retrieval and evaluation engineering, not model procurement.

XII. LIMITATIONS

Knowledge-graph residuals. 164 chunk-only entities still lack pericope-level MENTIONS in the graph itself (retrieval-side remap restores their visibility, the graph remains incomplete); alias coverage is $\sim 0.8\%$ of entities (dictionary and curated sources are exhausted; mass production requires entity-resolution merge-back); 67,306 relation candidates remain unclassified (the 77,953-pair pile minus the 10,647 Event-related pairs consumed or set aside by the P0 rescue); the event layer has only 26 native temporal/causal Event–Event edges live (the audit-time figure of 45 included miscounted person-succession edges and edges of since-deleted generic events) plus 277 unrescued Event–Event candidates; there is no coreference resolution; toneless-pinyin IDs can over-merge homophones; and duplicate-name splits (e.g., Israel as Place vs. Israelites as Group) await entity resolution.

Benchmark scale and head bias. 100 questions, 20 per category, asking predominantly head entities; per-category deltas below ≈ 0.05 on judged metrics are noise, and several architectural claims rest on question families of size 1–2, which we deliberately refuse to tune against.

Judged metrics. RAGAS scores carry ± 0.4 per-question noise at our scale and a noncommittal rule that penalizes honest refusal; the answer-coverage judge is a single local model. All retrieval claims are therefore anchored to deterministic metrics.

Generation. The deployed generator is an 8-bit E4B-class model; the attenuation analysis shows it discards part of the

retrieval gains. Conclusions about answer quality are conservative lower bounds.

Scope. One corpus, one language, one translation, one domain. The architecture’s signal set and dictionaries are domain-engineered; transfer requires re-instantiating those assets (though the methodological findings — ranking-layer gating, benchmark dependence, patch lifecycles — are domain-agnostic claims about system structure, not about the Bible).

XIII. FUTURE WORK

Evaluation first. Per Section XI, the next artifact is the benchmark, not the pipeline: (i) long-tail question sets targeting chunk-only entities and split-pericope person $\times$ event material; (ii) BRINK-style defect injection [10] — remove known edges, measure answer degradation — as a causal probe of KG value that no head benchmark can fake; (iii) population of the cross-chapter-aggregation and dual-anchor families to statistically meaningful sizes; (iv) judged-metric hygiene (non-committal exclusion, position/length debiasing [11]).

P1 — decoupled re-extraction. Feed extraction whole pericopes (storage unit $\neq$ extraction unit [40]); add a coreference pass scoped to pericope+chapter context following the LINK-KG recipe [9]; adopt true-loop gleanings as in the GraphRAG lineage [3] (e.g., the nano-graphrag implementation [41]); post-hoc entity resolution with lexical+embedding double signals and propose-then-confirm merging, writing merged surface forms back as aliases (the alias-generation mechanism identified in the cross-system survey); type assignment by majority vote; description regeneration by merge-and-summarize (cf. LightRAG [19]); ontology-grounded post-correction as the cheap alternative to re-extraction [28], [29]; and KG denoising with downstream verification [30].

P2 — the event-temporal layer. Promote the event layer from participants/places (now 84%/75.5%) to temporal-causal structure, following atomic-fact dual-time construction [34] and entity–event dual-graph designs [23], [24]; the 77,953-pair provenance-kept rejection pile and the 277 unrescued Event–Event candidates are the ready inputs.

P3 — aggregation retrieval. Community summaries (Leiden partitions [42] as in [3]) or RAPTOR-style summary trees [22] for cross-chapter aggregation; RAGK-style pre-entity re-retrieval [31] and cross-chunk graph augmentation [33] for cross-boundary relations.

Retrieval mechanics. Query decomposition for dual-anchor conjunctions (retrieve prophecy and fulfillment sides separately, then join on cross-reference edges); theme-level anchoring; and, if the strategy-prior fusion reaches its ceiling, calibrated score fusion in the spirit of [18] or graph-native retrieval scoring [20], [21].

Generation. Answer-model A/B under debiased judging, since the attenuation analysis locates the next bottleneck there.

XIV. CONCLUSION

We presented the complete architecture of a deployed GraphRAG system for Traditional Chinese Bible QA — from layout-geometric PDF parsing, pericope-centric chunking, grounded dual-track entity extraction, closed-schema relation extraction, and a three-source cross-reference network,

through a signal-adaptive six-route retrieval design terminated by cross-encoder reranking and linear rank fusion — and an unusually complete empirical record of optimizing it: a quantified audit tracing KG damage to granularity inheritance rather than chunk parameters; a completeness overhaul whose construction metrics all moved and whose benchmark reading was zero; the question-level diagnosis of that negative result; and a one-day ranking-layer remediation that surpassed all prior results (hit rate 0.97, EVENT 1.00, +0.16 over the semantic baseline), decomposed by ablation into complementary discrete and continuous mechanisms.

The system-level conclusions are, we believe, portable to any GraphRAG deployment. Graph assets have no value until the ranking layer transmits them; benchmark design decides which layer’s investment is visible at all; discrete patches have architecture-bound lifetimes; and the correct optimization order follows the measured bottleneck, not the data flow. In our deployment, honoring these rules delivered, in a day of work, a larger measured return than the deferred week-scale re-extraction plan was projected to yield on the same benchmark — and identified the next bottlenecks (benchmark design, then generation) before any further construction spend.

APPENDIX A STAGE-TO-CITATION MAP

Table XVI indexes, for every pipeline stage and experimental round, the literature it draws on and the role each work plays — so that each citation can be audited in place.

APPENDIX B REPRODUCIBILITY NOTES

Determinism and re-scoring. All retrieval metrics are computed programmatically against pericope-level gold references; every evaluation run archives per-question raw responses (sources, strategies, raw reranker and fused scores, route), and the metric suite can be recomputed bit-exactly from the archives. The fast retrieval-only loop was validated to agree bit-exactly with the full pipeline on all retrieval metrics.

Comparability across the three time points. Answer model (Gemma 4 E4B, 8-bit), judge model (Gemma 4 26B), decoding temperature (0.1), top- k (5), and route distribution (R1=20, R2=25, R3=21, R4=18, R5=10, R6=4, fallback=2) were verified identical; the entity-query strategy was off at the first two time points and on after the fusion round (its re-enablement is itself one of the audited repairs). The only other systematic difference between time points is the state of the knowledge graph and of the post-retrieval chain, as described in the text.

Rollbackability. Every P0 and fusion-round mutation is reversible: backfilled mention edges, co-occurrence relation edges, and TSK cross-reference edges carry provenance flags and can be deleted by a single predicate each; curated event nodes carry a source tag; alias injections have before-snapshots; and the rank-fusion layer, expansion caps, and entity-query strategy are configuration switches whose legacy settings reproduce the pre-remediation system exactly.

Scale of the evaluation. A full 100-question pipeline run takes ≈ 2.5 hours on the single-host deployment (36 min collection, 1 h 42 m RAGAS judging, 13 min coverage judging); the retrieval-only loop takes ≈ 15 minutes and was used for the α sweep and regression forensics.

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TABLE XVI
WHICH WORK IS CITED AT WHICH STAGE, AND WHY

Stage (section)	Works	Role of the citation
Problem framing (§I)	[1]–[4], [12]	faith-dimension LLM deficits motivating grounded QA; RAG foundations and surveys; GraphRAG paradigm
Chunking & granularity (§III-D, §VIII M5–M6)	[3], [16], [27], [34]	tokenizer/embedder fit; chunk size $\rightarrow$ extraction density (600 vs. 2,400 tokens $\approx 2\times$ entity references); finer units extract more completely; long-context degradation as the upper bound
Coreference & overlap (§VIII M4)	[6]	40.7% of relational facts cross-sentence, 17.6% need coreference — the structural ceiling of fragment-local extraction without coreference (used as <i>indirect</i> evidence; no peer-reviewed overlap-size quantification exists)
Entity extraction (§IV-A)	[7], [8], [17], [37]	Chinese segmentation/NER; the local classify-only LLM; structured prompting as noise suppression
Relation extraction (§IV-B)	[26], [32]	schema-based paradigm positioning; evidence-span grounding against context-divorced triples
Cross-references / TSK (§IV-C)	[5], [30]	public-domain cross-reference data; repair-by-reweighting rather than deletion
KG-quality remediation roadmap (§VIII, §XIII)	[9], [19], [28], [29], [31], [33], [40], [41]	coreference recipe (-45% duplication, length-amplified); post-hoc canonicalization / correction; cross-chunk reassembly; storage $\neq$ extraction unit; true-loop gleanings; merge-and-summarize descriptions
Retrieval substrate (§V-E)	[13]–[16]	dense retrieval; BM25; RRF hybrid; embedder and cross-encoder
Rank fusion (§V-F, §X-C)	[18]	calibrated graph/vector fusion as the sophisticated alternative to our single- α linear form
Evaluation framework (§VI)	[11], [35]	RAGAS metrics; judge-bias methodology (position/verbosity; noncommittal artifact)
Negative result & benchmark dependence (§IX, §XI)	[10], [11], [25]	KG-defect $\rightarrow$ answer non-linearity and defect injection; debiased gains; when graphs help RAG at all
Aggregation & event layer (future) (§IX-D, §XIII)	[3], [20]–[24], [34], [42]	RAPTOR trees / Leiden community summaries for cross-chapter aggregation; temporal-causal event modeling; graph-native retrieval scoring
Retrieval-side chunking sensitivity (§II)	[36]	cited <i>only</i> for retrieval-downstream chunking sensitivity — explicitly not extrapolated to KG construction

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